



UNIVERSITY OF CANBERRA

PROJECT FINAL REPORT

Project Title: Time Series Analysis of Stock Indices And Cryptocurrencies Using Econometric And Machine Learning Models Before And During The Covid-19 Pandemic

Project ID: 2026-S1-19

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1. Introduction

Financial markets produce large complicated time series data that can be affected by volatility, nonlinear, and surprising events from all around the world. In recent times, one of the major events was the coronavirus disease (COVID-19) pandemic that brought a lot of structural uncertainty and changes to both traditional stock markets and newly emerging cryptocurrency markets. This made financial prediction even harder than before and required robust predictive models.

The main task of this project is to analyse financial time series data using three different modelling techniques: ARIMA, GARCH, and LSTM which represents different views of financial behaviour about how markets behave; linearly and autocorrelated (ARIMA), where volatility Clustering is present and heteroscedasticity prevails (GARCH). And Finally, LSTM is capable of learning nonlinear patterns and long-term dependencies in sequential data. The main purpose is to see how models perform across different asset classes and market conditions.

From Yahoo Finance, the dataset was collected using the finance API in python code. For instance stock indices such as S&P 500, ASX 200, and Ibovespa, and cryptocurrencies like Bitcoin, Ethereum, and Dogecoin. These assets were selected to study both moderate and high volatility markets for a meaningful comparison across different financial conditions.

From project management perspective, the detailed structured planning done at the beginning was critical, with clear definition of roles and responsibilities between team members, continuous communication with sponsor and mentor, consistent follow up for any issues and effective risk management. Our team was working with the proposed plan, overcoming obstacles such as processing constraints and data alignment tweaks to ensure that the concise deliverable was reproducible, well documented and fit for use in practical scenarios.

This report provides an overview of how the project was executed, what deliverables were promised and delivered, and some sponsorship benefits realized.

1.1 Objectives and Purpose

The main objective of this project was to evaluate and compare the performance of ARIMA, GARCH and LSTM models across the different market conditions and financial assets.

Our objectives were:

- To collect and preprocess multi-asset financial time series data
- To implement ARIMA for modelling linear time dependencies
- To apply GARCH in capturing volatility clustering
- To develop LSTM models for nonlinear sequence learning
- To evaluate model performance using RMSE, MAE, and R^2
- To compare model behaviour across pre-COVID and COVID regimes

The other purpose of the project was to create a structured framework than can be reused and easy to implement in future research. It includes proper documentation, consistent evaluation metrics and valuable insights of results across various datasets.

This project also aimed to provide practical evaluation and reliability of various models under different market conditions.

1.2 Achievements

We successfully achieved the goals defined in the project proposal from this project. We successfully implemented all three models ARIMA, GARCH and LSTM. Each model was developed and tested using real financial data from yfinance, which allowed a realistic evaluation of their performance.

Another achievement was an organized data preprocessing pipeline was developed. This involved handling missing values, transforming price data into log-returns to make it stationarity and performing data normalization for deep learning algorithms. This step was essential in order to ensure consistency across models when applied to different datasets.

The project also provided a comprehensive comparison of model performance across different asset classes and market regimes. The results clearly showed how model performance varies depending on volatility and data characteristics. Moreover, the team successfully produced all required deliverables such as research report, project report and presentation materials.

2. Restatement of Scope and Requirements

The initial scope of the project was broad but was refined to a more workable scale during execution to ensure feasibility and better alignment with available resources. The final scope mainly focused on comparing three important models, ARIMA, GARCH and LSTM across stock indices and cryptocurrencies and analysing the performance of these models during pre-COVID and COVID periods.

Some aspects originally considered in the proposal, such as hybrid models and additional datasets, were not fully implemented due to time constraints and computational limitations. However, this helped the team concentrate on producing good results within the given timeframe.

The project required accurate data collection, proper preprocessing, implementation of forecasting models and evaluation using standard parameters like RMSE, MAE and R^2 . Finally, the project was completed with justified results, clear documentation and sponsor's requirements were also met. These outcomes were the key requirements during the submission of the project was consistently followed and met throughout.

3. Project Outcomes

The result of the project was a complete financial forecasting and analysis system. In this framework, data processing, model implementation and evaluation are organized and done in a way that is easy to adopt. What this framework does, is that it gives an organized form of the data pre-processing, model implementation, and evaluation to enable its easy adoption. This result is not only a set of results but also a repeatable system that can be used for future research as well.

The implementation of the models was a success, and the performance of these models was analysed for different financial assets and market conditions. The project provided valuable learnings on the effect of volatility and uncertainty on different models under inconsistent conditions. Overall, outcomes seem to align to initial objectives and also delivered satisfactory results for technical and project management aspects.

3.1 Deliverables

Deliverables	Description	Status
Project Proposal and Plan	Scope, Objectives, Methodology, Risks	Completed
Data integration	Yahoo Finance API data collection	Completed
Data Preprocessing	Cleaning	Completed
ARIMA Model	For time dependent linear patterns	Completed
GARCH Model	For time varying volatility	Completed
LSTM Model	For testing nonlinear sequential learning	Completed
Model Evaluation	Comparing RMSE, MSE and R2	Completed
Research Report	Academic paper with Abstract, Introduction, Related work, Results, Analysis and Conclusion	Completed
Poster	One-page A0 poster for visual summary of project structure and workflow, key findings	Completed
Final Project Report	Project Management documentation	Completed
Presentation	Visual project summary	Completed

3.2 Key Results

The results of this project highlight how the performance of forecasting models varies depending on asset type, volatility level, and market conditions. The comparison across stock indices and cryptocurrencies during both pre-COVID and COVID periods provides a deeper understanding of how each model behaves under different financial environments.

3.2.1 ARIMA Performance

ARIMA was the model that showed the highest performance across the majority of datasets. It was particularly most effective for stock indices such as the S&P 500, Ibovespa, and ASX 200 for both pre-COVID and COVID periods, with the model reaching an R-squared ranging from 0.88 to 0.99, showing that the model could capture most part of its variability, where price movements follow relatively structured and time-dependent patterns.

Figure 1 below shows the volatility plots with its corresponding prediction curves during pre-COVID period, for the best prediction (lowest RMSE and highest R-squared) among the indices: S&P 500, from the USA (Ticker: "^GSPC").

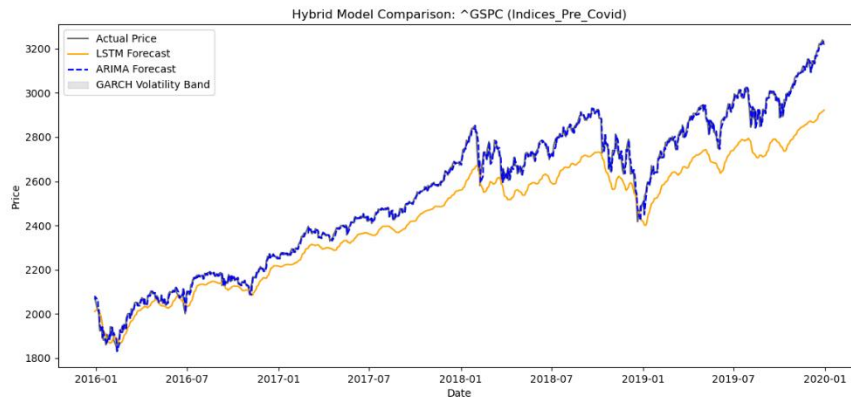


Figure 1. Volatility plots for the 20% training split along with their respective prediction curves for the S&P 500, from the USA (Ticker: "^GSPC") during pre-COVID times.

3.2.2 GARCH Performance

The GARCH model was used to capture volatility bands, especially in cryptocurrency markets. The results confirm the presence of volatility clustering, where big changes in price are followed by further fluctuations. This behaviour is more prominent in cryptocurrencies compared to stock indices. Figure 2 below shows Ethereum (Ticker: "ETH-USD") during pre-COVID times. As we can see, the volatility band for the American index practically does not exist, while for Ethereum, is wider and visible.

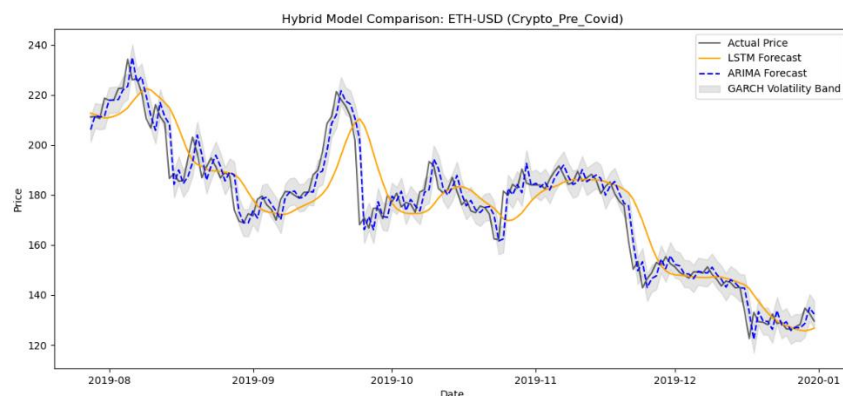


Figure 2. Volatility band (gray area) for the 20% training split along with their respective prediction curves for Ethereum (Ticker: "ETH-USD") during pre-COVID times.

3.2.3 LSTM Performance

The LSTM model, which is designed to capture nonlinear relationships, showed mixed performance across the datasets. Its effectiveness was limited by data size,

preprocessing, and computational constraints. Deep learning models require a huge amount of data and computational resources for training (Bergmann, n.d.).

In particular, LSTM forecast struggled during the COVID period (Figure 3) for the Australian stock indices, where sudden market changes reduced prediction accuracy, reaching the lowest R-squared from this group.

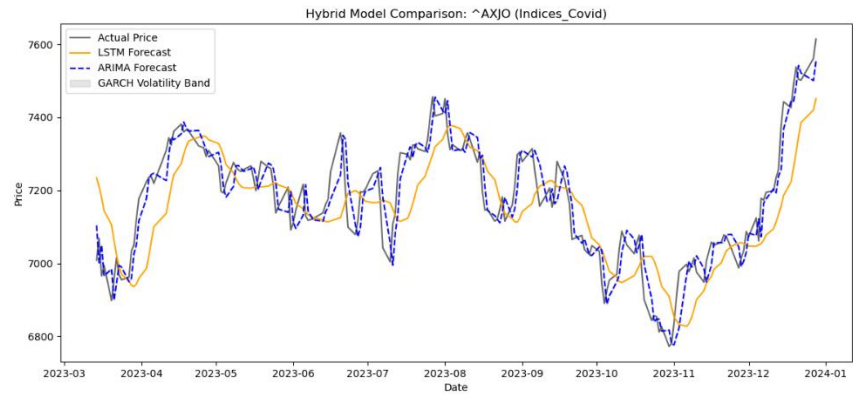


Figure 3. Volatility plots for the 20% training split along with their respective prediction curves for the ASX 200 from Australia (Ticker: "[^]AXJO") during COVID times.

3.2.4 Comparative Model Analysis

The results shown in Table 1 and 2 confirm that model performance is dependent on the nature of the data and market conditions.

ARIMA produced a better performance when compared to LSTM for both indices and cryptocurrencies in the studied periods of this work, showing the lowest RMSE and MSE values and highest R-squared.

ARIMA outperformed LSTM for both market indices and cryptocurrencies across the studied periods, delivering the lowest RMSE and MSE values and the highest R-squared values. This indicates that it was able to capture the main price movements of those assets. This finding is also consistent with findings from other researchers (Box et al., 2015) (Taslim & Murwantara, 2024). Another interesting fact is that the RMSE values from all ARIMA models increased for all datasets during COVID periods.

For, LSTM model, the results were mixed, with its respective RMSE values decreasing and increasing within the COVID period.

In contrast, the LSTM models delivered mixed results, with their respective RMSE values increasing and decreasing with COVID.

Table 1. RMSE, R-squared and MSE for ARIMA and LSTM models during pre-Covid and COVID periods for stock Indices.

Asset	Regime	RMSE_ARIMA	RMSE_LSTM	R2_ARIMA	R2_LSTM	MSE_ARIMA	MSE_LSTM
^GSPC	Indices_Pre_Covid	20.99	134.17	0.996	0.837	440.60	18,002.84
^GSPC	Indices_Covid	35.28	73.85	0.969	0.865	1,244.87	5,453.40
^BVSP	Indices_Pre_Covid	1,012.42	4,738.94	0.997	0.934	1.02E+06	2.25E+07
^BVSP	Indices_Covid	1,228.77	3,817.51	0.976	0.773	1.51E+06	1.46E+07
^AXJO	Indices_Pre_Covid	43.68	96.96	0.992	0.958	1,908.29	9,401.80
^AXJO	Indices_Covid	53.93	106.62	0.887	0.559	2,908.11	11,367.90

Table 2. RMSE, R-squared and MSE for ARIMA and LSTM models during pre-Covid and COVID periods for cryptocurrency.

Asset	Regime	RMSE_ARIMA	RMSE_LSTM	R2_ARIMA	R2_LSTM	MSE_ARIMA	MSE_LSTM
^GSPC	Indices_Pre_Covid	20.99	134.17	0.996	0.837	440.60	18,002.84
^GSPC	Indices_Covid	35.28	73.85	0.969	0.865	1,244.87	5,453.40
^BVSP	Indices_Pre_Covid	1,012.42	4,738.94	0.997	0.934	1.02E+06	2.25E+07
^BVSP	Indices_Covid	1,228.77	3,817.51	0.976	0.773	1.51E+06	1.46E+07
^AXJO	Indices_Pre_Covid	43.68	96.96	0.992	0.958	1,908.29	9,401.80
^AXJO	Indices_Covid	53.93	106.62	0.887	0.559	2,908.11	11,367.90

These findings suggest that no single model is superior to others. Selecting an appropriate model depends on asset type, volatility level, and market regime. Figures 1 to 3 and Table 1 and 2 support this comparative understanding.

3.3 Alignment with KPIs

A few of the proposal's key performance indicators (KPIs) was chosen in this project. Reproducibility, model performance, dataset reliability, and timely milestone fulfillment were given top priority. To provide dataset dependability, methods of preprocessing that were meticulously planned and implemented such as imputation of missing values and data transformation to log-returns were crucial. We were able to accurately compare and assess the ARIMA, GARCH, and LSTM models' performances by using RMSE, MSE, and R2.

In order to enable a repeatability factor, random generators were fixed within the code. The final code was posted to GitHub, which made it possible to maintain project files and implementations in a single location. Multiple runs were not conducted due to computing constraints, although the experimental setup and upkeep were constant.

All the major task such as data collection, modelling, assessment and reporting were completed as planned. Overall, the project was accomplished with good planning and execution by meeting its specified KPIs.

3.4 Quality Assurance Measure

In order to ensure that the results are reliable, consistent, and repeatable, quality assurance (QA) was quite helpful in this project. Data quality, model success, repeatability, and documentation according to the proposal's objectives were the main focuses of the QA procedure.

The team used well-defined preprocessing, well recognized assessment measures, and consistent modelling operations throughout the project to preserve quality. Summary of the main QA metrics and their outcomes are as follows:

QA Goal / Objective	Quality Measure Implemented	Evidence / Outcome
Dataset Reliability	Yahoo Finance API data was gathered and pre-processed (dropna, alignment, log-returns, scaling).	The data set became reliable and appropriate for time series modelling.
Data Consistency	Removed NA's	It made sure that all assets were fairly compared.
Model Performance	RMSE, MSE, and R2 are used to evaluate all models.	Performance comparison of ARIMA, GARCH, and LSTM
Reproducibility	The final version and fixed random seeds are available on GitHub.	Centralized project management and consistent outcomes
Model Stability	To prevent data leaking, use a time-based train-test split.	Increased forecasting outcomes' dependability
Documentation Quality	Project reports, organized research reports, and graphic results	Clearly stated approach and findings
Comparative Validation	Explicit description of the technique and findings Testing models in various market regimes and assets	Validated results under various volatility situations
Timeliness	Collecting milestones for the project from the proposal.	Every deliverable was finished on time.

The outputs from the project are reproducible, dependable and well documented because of QA procedures. We maintained high quality findings using the standard evaluation measure, consistent preprocessing and well-organized code management.

3.5 Impact and Value to Sponsor

The project delivers practical and scientific value to the sponsor through multiple major contributions.

- **Model Performance Insights:**

The project outlines a well-positioned contrast between ARIMA, GARCH, and LSTM models in various financial assets and market situations. Findings reveal that ARIMA is a stable performer for well-organized datasets, whereas GARCH efficiently models volatility trends, more so when it comes to crypto markets. Such revelations may be invaluable for forecasting model choices that depend on data properties.

- **Understanding of Market Behaviour:**

Looking at the pre-COVID and COVID times, research finds the reaction of various prediction techniques when exposed to not only stable but also very volatile markets. It also sheds light on how unexpected changes in market scenarios affect the accuracy of forecasting as well as the behaviour of volatility in financial markets.

- **Reproducible and Extendable Framework**

The project produces a prediction pipeline that is repeatable and backed by uniform model implementation and preprocessing. Using GitHub guarantees appropriate version control, which facilitates future project extensions and change tracking. This enables the sponsor to expand upon and repurpose the current structure.

- **Practical Application Value:**

By developing appropriate models for various asset kinds, the results can aid financial analysis. Additionally, the research lays the groundwork for upcoming advancements like real-time systems for forecasting or mixed models.

- **Alignment with Sponsor Objectives:**

The study is in line with the sponsor's goal of analysing and assessing economic projections methods. It provides a comprehensive and useful output, including models, findings, and documentation, that can facilitate additional study and real-world applications.

4. Resources

4.1 Team Contribution

We were able to complete the project as a team, cooperating and splitting work among various sections such as data collection, data cleaning, modelling, analysis and documentation. We allocated tasks given on each team member's abilities and specialties which not only helped increase efficiency but also made sure the workload was distributed evenly during the project.

All the team members were involved in both technical and non-technical aspects including programming, deployment of the model, analysing the model outcomes and preparing the report. Consistent meeting and discussion ensured smooth progress and facilitated the group to come up with solutions to each problem we faced. This team effort was crucial in completing the projects on time.

4.2 Revised Work Breakdown Structure

Task	Description	Responsible	Estimated Efforts
Project Initiation	Finalizing the project topic and scope, objectives	Team Member + Sponsor	50 hrs
Literature Review	Reviewing the financial time series studies and ARIMA, GARCH, LSTM	Team Member	100 hrs
Data Collection and Validation	Data used from Yahoo Finance	Team Member	70 hrs
Data Cleaning Preprocessing	Cleaning and transforming data	Team Member	110 hrs
Exploratory Data Analysis	Volatility patterns, pre covid and during covid, trend plots	Lead Developer	60 hrs
ARIMA Model	Implementing ARIMA model	Lead Developer + Programmer	85 hrs
GARCH Model	Implementing GARCH model	Lead Developer + Programmer	90 hrs
LSTM Model	Implementing LSTM model	Lead Developer + Programmer	110 hrs
Model Evaluation	Evaluating and comparing the models.	Team Member	80 hrs
Results discussion	Explaining the findings across indices and cryptocurrencies	Team Member	70 hrs
Poster	Preparing the poster of project	Documentation Lead	50 hrs
Final Report	Preparing final report	Documentation Lead + All members	100 hrs
Presentation	Presentation slides	Documentation Lead	30 hrs
Sponsor Feedback	Enhanced according to the feedback and requirements	Sponsor	40 hrs

The Work Breakdown Structure (WBS) and project hours outlined in the initial proposal were checked with the total amount of project work and turned out to be insufficient. The team then reconsidered the planning and set more sufficient research time, processing time, planning of model, testing model, model reporting and project improvement in order to achieve better results.

So, for the final report, the WBS was reconsidered using deliverable based decomposition approach. With this approach the project was segmented into smaller

work packages linked to the main project outputs, including research design, dataset preparation, ARIMA modelling, GARCH modelling, LSTM modelling, evaluation, reporting, and presentation. This aligned with PMI guidance, which involves a WBS being used to structure the overall scope of project work into work packages.

4.3 System Setup

Our project was developed using Python language and its associated libraries. We used finance library to integrate the financial data from Yahoo Finance via an API utilising the Python finance API which withdraws data from the Yahoo Finance website. We used key libraries such as Pandas and NumPy for data handling, Statsmodels and Arch package for statistical modelling for ARIMA and GARCH and for LSTM model we used Tensorflow/ keras library.

This system was developed on a standard laptop without dedicated GPU resources. This was very much capable of handling the models. But this had a limitation for LSTM model as it was taking longer computation time and model tuning.

5. Challenges and Learning

During this research project phase, we got several technical and project management challenges. This problem helped us to learn more about our analytical skills as well technical knowledge and to work more as a team.

Challenge	Impact	Learning Outcomes
LSTM Model complexity	Higher computational power and tuning compared to ARIMA & GARCH model.	Deep learning models need sufficient resources and careful tuning.
Time Constraints	Limited ability to perform extensive hyperparameter tuning and multiple runs	Importance of prioritizing the key task and focusing on core project objectives
Computational Limitation	Slow training process for LSTM	Optimizing the model setup and manage the resources efficiently.
Academic Tone	Used strong wording	Must be soft tone and data specific
Balancing Documentation and Coding	Managing both technical work and report writing simultaneously	Improved time management and coordination within the team.

5.1 Risk Mitigated

Risk Identified	Mitigation Strategy	Outcome
Missing Data and gaps	Evaluate missingness of dataset and use of Dropna function	Cleaned dataset without NA's.

Model Overfitting	Use of time-based train test split	Improved model reliability & avoided data leakage
Time Constraints	Task priority and focus on core objectives	Completed project within the timeline
Reproducibility Risk	Use of fixed random seeds	Consistent results

6. Handover Material

The project materials are organised and archived in such a way that it can be handed over to the sponsors, and they can reuse, edit and further build upon any time. We have created a research report, poster, source code, and presentation, which will make it easier for them to review and complete more on those results.

6.1 Usage Instructions and Maintenance Tips

The user can run the Python code to access this project. The dataset can easily be loaded using the yfinance API. Always use the same tickers when comparing to existing results. Users can change the models or edit the code for their desired results. Check for data availability and calendar difference before modelling when adding new assets.

6.2 Known Limitations

- The project is academic research in forecasting comparison, not a trading or investment system.
- Due to limited computational resources LSTM models training window are longer.
- Only 3 stock indices and cryptocurrencies were used so won't be applicable for all others indices and cryptocurrencies.

7. Future Work and Opportunities

There are a number of ways that this project might be expanded and enhanced well past its initial stage. One of the major pathways is the designing of hybrid models that merge ARIMA, GARCH, and LSTM, thereby exploiting the advantages of each method. Such a move could bring better forecast accuracy as it would consider linear components, changing volatility, and nonlinear features all at once, and would be a natural continuation of the work that has been done on the models in this project.

Besides, integrating external factors like macroeconomic indicators, financial news, or sentiment analysis is also very significant. As financial markets are impacted by many external variables, including these elements may not only help models become more sensitive to sudden market changes but also help to better reflect changes as threat periods, such as COVID-19, of high market volatility are attaining. This would make the present structure a more reasonable and useful forecasting system.

From a practical and project standpoint, upcoming work can be directed towards enhancing the performance of LSTM by better hyperparameter tuning, larger data sets, and more computational resources. Besides that, the project can be developed into a real-time forecasting system and broadened to cover more types of financial assets. Such upgrades will not only make the final products more user-friendly but also give the sponsor an opportunity to turn the project into a more sophisticated and scalable solution.

8. Conclusion

This project completely created a financial forecasting system that consists of data pretreatment, model setup, and comparison of different financial assets and market situations. Besides research report, the modelling pipeline, and documentation, all key deliverables were completed and ready for real handover, and project results were in line with the initial plan. Study results indicated that GARCH gave deep understanding of volatility movements mainly in the crypto markets whereas ARIMA was the best reliable performer of financial time series.

The LSTM model was a good start, but it needed more development and additional processing power for stable performance. These findings have demonstrated the necessity of choosing forecasting models considering data features as well as prevailing market conditions. From a project management perspective, there was good demonstration of project planning, coordination, and risk management.

Resolving issues related to data alignment, computing limits, and time constraints effectively resulted in timely completion of all objectives. All in all, the project gives a setup which can be reproduced and extended that besides giving the sponsor practical value also offers analysis, thereby forming a firm base for further development.

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APPENDIX

Project Code link on GitHub:

https://github.com/Rojanpun/Final_Project